

## Whole-Number Division: Up to Four Digits by One-Digit Divisors Answer Key

Grades 4–5 • Fluency Practice

### Quick Answers

---

|        |        |        |         |
|--------|--------|--------|---------|
| 1. 42  | 4. 400 | 7. 297 | 10. 545 |
| 2. 75  | 5. 405 | 8. 858 | 11. 5   |
| 3. 103 | 6. 129 | 9. 609 | 12. 66  |

---

### Common wrong answers

---

3. *If they got 13: skipped the empty tens place instead of writing a 0 in the quotient, giving 13*  
5. *If they got 45: skipped the empty tens place and wrote a two-digit quotient of 45*  
6. *If they got 129.1: wrote the leftover as a decimal digit (129.1) instead of a remainder*  
9. *If they got 69: dropped the 0 from the tens place and wrote 69*  
10. *If they got 546: rounded the leftover up to one more box instead of ignoring it*  
11. *If they got 545: reported the quotient instead of the leftover the question asked for*  
12. *If they got 594: stopped after the first division and gave the tickets per class*
- 

**1.**  $168 \div 4$ . Divide from the left. 1 hundred is smaller than 4, so start with 16 tens:  $16 \div 4 = 4$  tens. Multiply and subtract:  $16 - 16 = 0$ . Bring down the 8:  $8 \div 4 = 2$  ones. The quotient is 42.  
. Check:  $42 \times 4 = 168$ .

**2.**  $375 \div 5$ . 3 hundreds is smaller than 5, so start with 37 tens:  $37 \div 5 = 7$  tens with 2 left ( $5 \times 7 = 35$ ,  $37 - 35 = 2$ ). Bring down the 5 to make 25:  $25 \div 5 = 5$  ones. The quotient is 75.  
. Check:  $75 \times 5 = 375$ .

**3.**  $618 \div 6$ .  $6 \div 6 = 1$  hundred;  $6 - 6 = 0$ . Bring down the 1: 1 is smaller than 6, so write **0** in the tens place and bring down the next digit. That makes 18:  $18 \div 6 = 3$  ones. The quotient is 103.  
. The 0 is not optional — skipping the place gives 13, which is far too small, since  $13 \times 6 = 78$ .

**4.** Estimating  $2,835 \div 7$  with the compatible number 2,800:  $28 \div 7 = 4$ , so  $2,800 \div 7 = 400$ . Nadia's estimate is 400.  
. This also tells her the exact quotient will have 3 digits.

**5.**  $2,835 \div 7$ . 28 hundreds  $\div 7 = 4$  hundreds;  $28 - 28 = 0$ . Bring down the 3: 3 is smaller than 7, so write **0** in the tens place. Bring down the 5 to make 35:  $35 \div 7 = 5$  ones. The quotient is 405

, which is very close to the estimate of 400 from problem 4 — a good sign the work is right.

*Watch for:* a two-digit answer of 45. That is the empty tens place skipped instead of recorded, and the estimate of 400 rules it out at a glance.

**6.**  $517 \div 4$ .  $5 \div 4 = 1$  hundred with 1 left. Bring down the 1 to make 11:  $11 \div 4 = 2$  tens with 3 left ( $4 \times 2 = 8$ ,  $11 - 8 = 3$ ). Bring down the 7 to make 37:  $37 \div 4 = 9$  ones with 1 left ( $4 \times 9 = 36$ ,  $37 - 36 = 1$ ). Since 1 is smaller than 4, it stays as the remainder: 129 R 1. Check:  $4 \times 129 + 1 = 516 + 1 = 517$ .

*Watch for:* an answer written as 129.1. That is the remainder pasted on as a decimal digit, which is not what it means — the leftover 1 is 1 out of 4, not one tenth.

**7.**  $891 \div 3$ .  $8 \div 3 = 2$  hundreds with 2 left. Bring down the 9 to make 29:  $29 \div 3 = 9$  tens with 2 left. Bring down the 1 to make 21:  $21 \div 3 = 7$  ones exactly. The quotient is 297. Check:  $297 \times 3 = 891$ .

**8.** The missing number is the dividend. Division and multiplication undo each other, so if  $\square \div 6 = 143$ , then  $\square = 143 \times 6$ . Multiply:  $143 \times 6 = 858$ . The number is 858. Check by dividing back:  $858 \div 6 = 143$ .

**9.**  $4,872 \div 8$ . 4 thousands is smaller than 8, so start with 48 hundreds:  $48 \div 8 = 6$  hundreds;  $48 - 48 = 0$ . Bring down the 7: 7 is smaller than 8, so write **0** in the tens place. Bring down the 2 to make 72:  $72 \div 8 = 9$  ones. The quotient is 609. Check:  $609 \times 8 = 4,872$ .

*Watch for:* 69 — the same dropped zero as problem 3, one place further left.

**10.** Divide the pencils into groups of 6:  $3,275 \div 6$ . 32 hundreds  $\div 6 = 5$  hundreds with 2 left ( $6 \times 5 = 30$ ). Bring down the 7 to make 27:  $27 \div 6 = 4$  tens with 3 left. Bring down the 5 to make 35:  $35 \div 6 = 5$  ones with 5 left. So  $3,275 \div 6 = 545 \text{ R } 5$ . The question asks only for *full* boxes, so the remainder is ignored here: 545 full boxes.

*Watch for:* 546. Rounding the leftover up is right for “how many boxes are needed”, but there are only 5 pencils left, not a full box of 6.

**11.** Same division:  $3,275 \div 6 = 545 \text{ R } 5$ . This time the question asks what is left after every full box is packed, and that is exactly the remainder: 5 pencils. Check with the division equation:  $6 \times 545 + 5 = 3,270 + 5 = 3,275$ .

*Teaching note:* problems 10 and 11 are the same division. What changes is what the leftover means — that decision is the skill being practised. A student who writes 545 here has answered problem 10 twice.

**12.** Two divisions, one after the other.

**Step 1 — tickets per class:**  $4,158 \div 7$ . 41 hundreds  $\div 7 = 5$  hundreds with 6 left ( $7 \times 5 = 35$ ). Bring down the 5 to make 65:  $65 \div 7 = 9$  tens with 2 left. Bring down the 8 to make 28:  $28 \div 7 = 4$  ones. Each class gets 594 tickets.

**Step 2 — tickets per student:**  $594 \div 9$ . 5 is smaller than 9, so start with 59:  $59 \div 9 = 6$  tens with 5 left ( $9 \times 6 = 54$ ). Bring down the 4 to make 54:  $54 \div 9 = 6$  ones. Each student receives 

|    |
|----|
| 66 |
|----|

 tickets.

*Estimate check:* 4,158 is close to 4,200 and  $4,200 \div 7 = 600$ , so about 600 per class;  $600 \div 9$  is between 60 and 70. The answer 66 fits.